Local-resolvent formulation and assessment of linear-scaling Kohn–Sham calculations

Albert Linda[1, *]

[1] *Department of Materials Science and Engineering, Indian Institute of Technology Kanpur, Kanpur 208016, Uttar Pradesh, India*

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Kohn–Sham density-functional theory is expensive because it needs orbitals that extend over the whole system. For insulators this can be avoided by using only the diagonal of the one-electron resolvent, $g(\boldsymbol{r}, z) = \langle \boldsymbol{r}|(z - \hat{h})^{-1}|\boldsymbol{r}\rangle$, at a few complex energies z. We call the scheme local-resolvent density-functional theory (LRes-DFT). The density and the band energy are linear in g, so neither orbitals nor a kinetic-energy functional are needed. The complex energies are M equally spaced points on a circle that encloses the occupied levels, offset by half a spacing so that no point lies on the real axis. Summing the resolvent over them is equivalent to occupying each level with $1/(1 + u^M)$, where u is the level's energy measured from the circle's center in units of its radius. This is a smooth step that is near 1 inside the circle and near 0 outside, and it stays between 0 and 1 for every M. If the circle moves with the chemical potential, forces at fixed electron number are exact. Gauss–Legendre and Zolotarev rules need fewer points when the gap is small. In three dimensions we use eigenpairs of small windows around each molecule instead of resolvents. This is divide-and-conquer (DC) with a different occupation function, and it is as accurate as DC with Fermi–Dirac occupations. For H_2 molecular crystals of up to 1024 electrons it agrees with Kohn–Sham to 2.5×10^{-7} Ha per electron. On one core it is slower than dense diagonalization below about 500 electrons and 3.5 times faster at 1024 electrons. It loses accuracy for disordered or compressed systems and for diffuse basis functions.

I. INTRODUCTION

Density-functional theory (DFT) [1, 2] is the standard tool for computing the energetics of materials. In materials problems the cost of Kohn–Sham (KS) calculations is often the limiting factor. This includes generalized stacking-fault energies of alloys [3], solute segregation at grain boundaries [4], and large charge-density data sets [5]. The KS kinetic energy is computed from $N/2$ orbitals that extend over the whole cell, so storage grows as $\mathcal{O}(N^2)$ and diagonalization or orthogonalization as $\mathcal{O}(N^3)$ with the number of electrons N. For insulators, in which the density matrix is local, this cost can be reduced to $\mathcal{O}(N)$. The present work is restricted to such systems: gapped molecular solids, including disordered ones. Metals, which are central to the applications above, require finite electronic temperature and are not treated here.

Two routes to lower cost are well established. Orbital-free DFT [6–8] keeps only the density and approximates the kinetic energy. Linear-scaling KS methods keep the exact kinetic energy and use the locality (nearsightedness) of the density matrix [9–12]. They include divide-and-conquer (DC) [13] and fragment methods [14], density-matrix minimization [15, 16], localized-orbital methods [17, 18], Fermi-operator and Chebyshev expansions [11, 19], Krylov-subspace methods [20], recursion [21] and stochastic DFT [22].

A third family uses Green's functions on a complex energy contour. Korringa–Kohn–Rostoker (KKR) methods integrate on a complex contour as a matter of course [23], and the LSMS, LSGF and tight-binding KKR variants compute site-diagonal Green's functions in local zones [24–26]. Contour-integral forms of the Fermi operator [27, 28] and pole expansions of the Fermi–Dirac function [29–32] compute the density from a few resolvents. With selected inversion, pole expansions scale as $\mathcal{O}(N)$, $\mathcal{O}(N^{1.5})$ and $\mathcal{O}(N^2)$ in one, two and three dimensions [33]. The same rational filters also underlie contour-integral eigensolvers [34–37]. Section VII compares these approaches with the present one.

The formulation used here belongs to this third family. Its starting point, the KS density as a contour integral of the diagonal resolvent, is not new. We call the resulting scheme local-resolvent DFT (LRes-DFT). We study one simple realization in detail, with proofs and measurements, and report what it delivers and what it does not. The paper does four things.

1. It gives the formulation and its limits: which quantities are linear in g, why the diagonal resolvent exists only for a discretized Hamiltonian, and which operators it cannot represent.

2. It analyzes the half-step trapezoid rule on a circle (the Butterworth filter of contour-integral eigensolvers [36]) as a KS occupation function. The occupation $1/(1 + u^M)$ always lies in $[0, 1]$, the error obeys an a priori bound, and a rigidly moving circle gives exact fixed-N forces at every M. Gauss–Legendre and Zolotarev rules need fewer poles, and we count how many.

3. It evaluates the occupation in local windows in three dimensions, which is a DC method. We compare it, in self-consistent calculations of up to 1024 electrons, with DC using Fermi–Dirac occupations,

* albert.hzbn@gmail.com

Table 1. Main symbols.

Symbol	Meaning
$g(\boldsymbol{r}, z)$	local resolvent, Equation (1)
z_j, w_j	nodes and weights on the circle
c, R	center and radius of the circle
μ, E_{lo}	chemical potential; lower edge of the circle
M	number of complex energies
u, $r_M(u)$	$(\varepsilon - c)/R$; occupation $1/(1+u^M)$
Δ, W_{occ}	gap and occupied band width
R_w	window radius (3D)
N, N_b	electrons; basis functions

with dense KS diagonalization (the fastest of three LAPACK paths) and with a warm-started iterative KS solver.

4. It reports the cases in which the method is inaccurate.

We checked the analytic identities with computer algebra on explicit instances. Code and data are available [38].

In brief, the half-step rule works as designed in one dimension. In three dimensions the method reduces to divide-and-conquer, it agrees with Kohn–Sham to about 10^{-7} Ha per electron for gapped molecular crystals, and it is faster than dense diagonalization only above roughly 500 electrons.

The development proceeds as follows. Section II gives the formulation, with proofs in Appendix A. Section III describes the algorithms and Section IV the computational details. Sections V and VI present results in one and three dimensions, Section VII relates the method to other approaches, and Section VIII summarizes the limitations.

II. FORMULATION

Table 1 lists the main symbols used below.

A. The local resolvent

We consider spin-unpolarized electrons in an effective one-electron Hamiltonian $\hat{h} = -\frac{1}{2}\nabla^2 + v(\boldsymbol{r})$, with $v = v_{\mathrm{ext}} + v_{\mathrm{H}}[n] + v_{\mathrm{xc}}[n]$ the external, Hartree and exchange–correlation (xc) potentials (Hartree atomic units). Throughout, $\hat{h}$ is the Hamiltonian discretized on a grid or in a basis, i.e., a finite Hermitian matrix with eigenvalues ε_i and orthonormal eigenvectors ϕ_i. On a grid with volume element ΔV we use orthonormal grid vectors $|\boldsymbol{r}\rangle$, so that $|\phi_i(\boldsymbol{r})|^2$ is a probability per grid point and $n(\boldsymbol{r}) = 2\sum_{i\,\mathrm{occ}} |\phi_i(\boldsymbol{r})|^2/\Delta V$. The local resolvent is

$$g(\boldsymbol{r}, z) = \langle \boldsymbol{r}|(z - \hat{h})^{-1}|\boldsymbol{r}\rangle = \sum_i \frac{|\phi_i(\boldsymbol{r})|^2}{z - \varepsilon_i}, \qquad (1)$$

for complex z away from the eigenvalues. The eigenvectors appear here only to explain what g is; the algorithms do not compute global eigenvectors.

Remark 1 (Continuum limit). Everything below is for a discretized $\hat{h}$. In the continuum the diagonal of the resolvent is finite in one dimension but not in three: the free Green's function $G_0(\boldsymbol{r}, \boldsymbol{r}'; z) = -\mathrm{e}^{\mathrm{i}k|\boldsymbol{r}-\boldsymbol{r}'|}/(2\pi|\boldsymbol{r} - \boldsymbol{r}'|)$, $k = \sqrt{2z}$, diverges as $\boldsymbol{r}' \to \boldsymbol{r}$, and on a grid $g(\boldsymbol{r}, z)$ grows as $1/\Delta x$. The divergent term, $-1/(2\pi|\boldsymbol{r}-\boldsymbol{r}'|)$, does not depend on z, so it drops out of every contour integral of g over a closed contour, and the density and band energy below remain finite. The local resolvent itself is defined only for a discretized Hamiltonian.

B. What is linear in g

Proposition 1. *Let C be a closed contour, traversed counterclockwise, that encloses the occupied eigenvalues and no others. Then (a) $n(\boldsymbol{r})\Delta V = 2 \cdot \frac{1}{2\pi\mathrm{i}} \oint_C g(\boldsymbol{r}, z)\, dz$, and (b) $E_{\mathrm{band}} \equiv 2\sum_{i\,\mathrm{occ}} \varepsilon_i = 2 \cdot \frac{1}{2\pi\mathrm{i}} \oint_C z \sum_{\boldsymbol{r}} g(\boldsymbol{r}, z)\, dz$.*

The KS kinetic energy follows as $T_s = E_{\mathrm{band}} - \sum_{\boldsymbol{r}} v(\boldsymbol{r}) n(\boldsymbol{r}) \Delta V$. The kinetic energy T_s is not linear in g, but it follows from g without approximation through E_{band} and n, and the total energy has the usual KS form,

$$E = E_{\mathrm{band}} - \textstyle\sum_{\boldsymbol{r}} v_{\mathrm{Hxc}}(\boldsymbol{r})\, n(\boldsymbol{r})\, \Delta V + E_{\mathrm{H}}[n] + E_{\mathrm{xc}}[n] + E_{nn}, \qquad (2)$$

with $v_{\mathrm{Hxc}} = v_{\mathrm{H}} + v_{\mathrm{xc}}$ and E_{nn} the nuclear repulsion.

Remark 2 (Scope). The diagonal g on a contour that encloses the whole spectrum contains far more information than the density. For a grid Hamiltonian with a local potential, the large-contour moment $\frac{1}{2\pi\mathrm{i}} \oint_{|z|=R'} z\, g(\boldsymbol{r}, z)\, dz$ (R' larger than the spectral radius) equals $h_{\boldsymbol{rr}} = t_{\boldsymbol{rr}} + v(\boldsymbol{r})$, where t is the discretized kinetic operator. This identity fails in a non-orthogonal basis and with nonlocal pseudopotentials.

Some quantities need off-diagonal elements of the density matrix: nonlocal pseudopotential energies, exact exchange and meta-GGA kinetic-energy densities. These are not given by g, so the formulation as stated is limited to local potentials and semilocal functionals.

C. The half-step trapezoid rule

We take C to be a circle that encloses the occupied levels and excludes the empty ones. It passes through a point E_{lo} below the lowest eigenvalue and through the chemical potential μ, which lies in the gap. Its center and radius are $c = (\mu + E_{\mathrm{lo}})/2$ and $R = (\mu - E_{\mathrm{lo}})/2$. We discretize it with an even number M of points at half-step angles,

$$z_j = c + R\zeta_j, \qquad \zeta_j = \mathrm{e}^{\mathrm{i}\pi(2j+1)/M}, \qquad w_j = \frac{R\zeta_j}{M}, \quad (3)$$

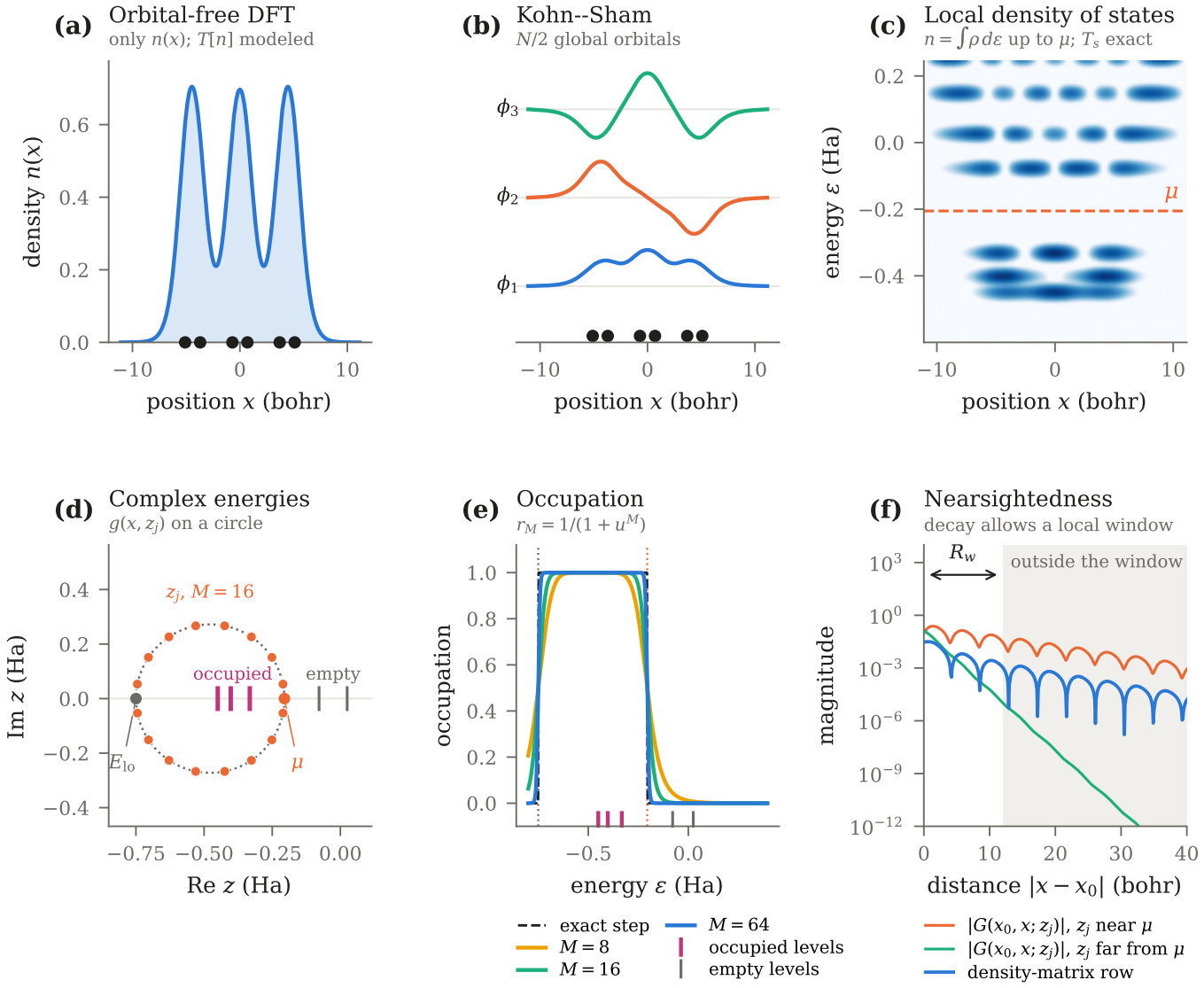


Figure 1. Formulation (symbols are defined in Section II), shown for a one-dimensional $(H_2)_3$ chain (black dots: nuclei) in panels (a)–(e) and an $(H_2)_{24}$ chain in panel (f). (a) Orbital-free DFT keeps only the density $n(x)$ and models the kinetic energy. (b) KS DFT computes $N/2$ orbitals ϕ_i that extend over the whole system. (c) Local density of states $\rho(x,\varepsilon)$ (color map, logarithmic scale). The density is its integral up to the chemical potential μ, and $g(x,z)$ is its transform to complex energies. (d) g is sampled at M complex energies z_j (orange) on a circle through E_{lo} and μ that encloses the occupied levels (magenta ticks) and excludes the empty ones (gray ticks); the axes are to scale. (e) The half-step trapezoid rule gives the occupation $r_M = 1/(1+u^M)$, which equals $1/2$ at E_{lo} and at μ; the ticks at the bottom mark the occupied (magenta) and empty (gray) levels of the chain. (f) The resolvent $G(x_0, x; z_j)$ and the density matrix decay exponentially with distance in this gapped system, so the density in a region can be computed from a window of radius R_w (the shaded part lies outside the window).

$j = 0, \ldots, M-1$. We call the z_j nodes; in the pole-expansion literature they are called poles, and the cost is counted in complex solves. The ζ_j are the roots of $\zeta^M = -1$, and w_j are the trapezoid weights of $\frac{1}{2\pi \mathrm{i}} \oint dz$ (the factor $1/2\pi\mathrm{i}$ is included). The nodes come in complex-conjugate pairs, so only $M/2$ resolvents are needed for a real symmetric $\hat{h}$.

Summing the resolvent over the nodes can be done in closed form, which gives the occupation function directly.

Lemma 1 (Closed form)**.** $\frac{1}{M}\sum_j \zeta_j/(\zeta_j - u) = 1/(1+u^M)$ *for all complex* $u \neq \zeta_j$*. Hence*

$$\hat{P}_M \equiv \sum_j w_j (z_j - \hat{h})^{-1} = \left[1 + \left(\frac{\hat{h} - c}{R}\right)^M\right]^{-1}, \qquad (4)$$

and in a non-orthogonal basis with overlap $S > 0$*,* $\sum_j w_j (z_j S - H)^{-1} = r_M(S^{-1}H)S^{-1}$*.*

The approximate density is $n_M(\boldsymbol{r})\Delta V = 2[\hat{P}_M]_{\boldsymbol{rr}}$, and

each eigenstate is occupied with weight

$$r_M(u) = \frac{1}{1+u^M}, \qquad u = \frac{\varepsilon - c}{R}. \tag{5}$$

For $M = 16$, a state with $u = 0.8$ receives 0.973 and a state with $u = 1.25$ receives 0.027; for $M = 64$ these become $1 - 6.3 \times 10^{-7}$ and 6.3×10^{-7}. This filter is not new: it is the Butterworth filter that the half-step trapezoid rule on a circle gives in contour-integral eigensolvers [36, 37]. What we add is its use as the occupation function of a KS calculation, with the consequences derived below.

Because the occupation is an explicit function, its range and its error for a single state can be bounded.

Lemma 2 (Occupations and error per state)**.** *For even M and real u, $r_M(u) = 1/[1 + \exp(M \ln|u|)]$, so $0 < r_M \le 1$. Hence $0 \preceq \hat{P}_M \preceq 1$, and $\hat{P}_M$ is a valid (ensemble-representable) one-body density matrix [39]. If $|u| \le \rho < 1$, then $1 - r_M \le \rho^M$; if $|u| \ge 1/\rho$, then $r_M \le \rho^M$.*

Two remarks on this rule. First, the ordinary trapezoid placement ($\zeta^M = +1$) gives $1/(1-u^M)$, which exceeds 1 inside the circle and is negative outside; the half-step placement avoids this. Second, unlike a Fermi–Dirac function, r_M is not monotonic in ε: it increases for $\varepsilon < c$ and decreases for $\varepsilon > c$, so a state below $E_{\rm lo}$ is occupied by less than 1/2. No concave entropy-like function generates r_M, so the finite-temperature (Mermin) minimum principle [40] has no direct analog here. Consequently we do not claim that the energy is minimal. We show only that it is stationary, which is what forces require.

D. Stationarity and forces

We next show that n_M is the exact derivative of a functional of the potential. This gives stationarity of the energy and exact forces.

Proposition 2. *Let $\Omega_M[v] = \mathrm{Tr}\, W_M(\hat{h}[v])$ with $W_M(\varepsilon) = -2\sum_j w_j \ln(z_j - \varepsilon)$. W_M is analytic for real ε because no node lies on the real axis, and real because the nodes and weights come in conjugate pairs; $W_M' = 2r_M$. For fixed c and R, $\partial\Omega_M/\partial v(\boldsymbol{r}) = n_M(\boldsymbol{r})\Delta V$ exactly.*

This identity holds for any smooth occupation function, including the Gauss–Legendre and Zolotarev rules below. Its consequence is that the self-consistent solution is a stationary point of

$$G_M[v] = \Omega_M[v] - \textstyle\sum_{\boldsymbol{r}}(v - v_{\rm ext})\, n_M[v]\, \Delta V + E_{\rm H}[n_M[v]] + E_{\rm xc}[n_M[v]] + E_{nn} \tag{6}$$

at fixed c and R. Hence the Hellmann–Feynman force (the derivative of $\hat{h}$ at fixed density) is the exact derivative of G_M (Appendix A). Here Ω_M plays the role of a grand potential and G_M adds the double-counting terms, so this is a grand-canonical statement. To conserve the electron number, the circle is translated along the energy axis together with μ, with its radius unchanged, so that $E_{\rm lo}$ moves with μ. The functional $A_M = G_M + \mu N$ then has N as a constraint:

Proposition 3. *Keep R fixed and set $c = \mu - R$. Then $\partial\Omega_M/\partial\mu = -N_M$, with $N_M = \mathrm{Tr}\, 2\hat{P}_M$, and $A_M = G_M + \mu N$ is stationary with respect to v and μ at the self-consistent solution with $N_M = N$. The Hellmann–Feynman force equals $-dA_M/dX$ exactly at every M.*

If $E_{\rm lo}$ is held fixed instead, the radius R changes with μ. The fixed-N energy, Equation (2), is then stationary only up to terms that vanish as the occupation error vanishes. Forces are correct only approximately at small M (Section V D).

E. Error bound

The density error of the half-step rule falls as ρ^M, where $\rho < 1$ measures how close the worst state lies to the circle. Here ρ is the largest of three numbers. They measure how far $|u|$ of the lowest level, of the highest occupied level and of the lowest empty level lies from the circle $|u| = 1$ (occupied levels must have $|u| < 1$, empty levels $|u| > 1$). The precise statement is:

Theorem 1. *Let $E_{\min}$ be the lowest eigenvalue, $E_{\rm lo} = E_{\min} - \delta$, $\Delta_- = \mu - \varepsilon_{\rm HOMO}$ and $\Delta_+ = \varepsilon_{\rm LUMO} - \mu$, where HOMO and LUMO denote the highest occupied and lowest unoccupied states, and let*

$$\rho = \max\left(\left|1 - \frac{\delta}{R}\right|,\ 1 - \frac{\Delta_-}{R},\ \frac{1}{1+\Delta_+/R}\right) < 1. \tag{7}$$

Then $\sum_{\boldsymbol{r}} |n_M - n|\Delta V \le 2(N_{\rm occ} + N_{\rm near})\rho^M + 2N_g 2^{-M}$, where $N_{\rm occ} = N/2$, $N_{\rm near}$ is the number of empty states with $1 < |u| < 2$, and N_g is the dimension of $\hat{h}$.

Here $2R = W_{\rm occ} + \Delta_- + \delta$, with $W_{\rm occ} = \varepsilon_{\rm HOMO} - E_{\min}$ the occupied band width. The rate does not depend on the largest eigenvalue of $\hat{h}$, which grows as $1/\Delta x^2$ on a grid; the bound depends on N_g only logarithmically, through $2N_g 2^{-M}$. The bound also shows the weakness of the rule. For small gaps $\ln(1/\rho) \approx \Delta/(2R)$, so M grows as $(W_{\rm occ}/\Delta)\ln(1/\epsilon)$ for a target accuracy ϵ, whereas optimal rational approximations need $\mathcal{O}[\ln(W/\Delta)\ln(1/\epsilon)]$ poles, with W the full spectral width [29, 31, 36]. The theorem applies to the 1D implementation; for the 3D windows there is no a priori bound, and we measure the window error instead.

F. Cost

In one dimension, a three-point finite-difference (FD) Hamiltonian is tridiagonal, and the diagonal of its resolvent can be computed exactly in $\mathcal{O}(N_g)$ operations:

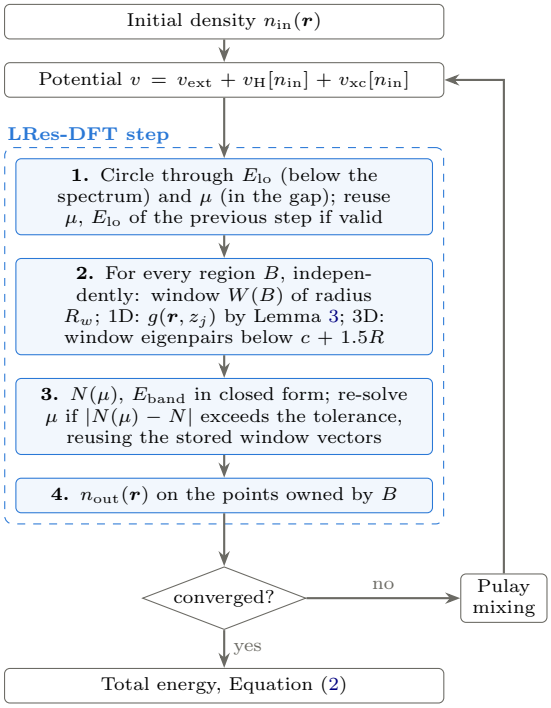


Figure 2. SCF cycle of LRes-DFT. Only the dashed box differs from KS DFT, where the same step diagonalizes the full Hamiltonian. Each region is computed from its own window, so the work grows with the number of regions.

Lemma 3 (Two-sided recursion)**.** *For tridiagonal $A = z - \hat{h}$ with diagonal a_i and off-diagonal b_i, let $L_1 = a_1$, $L_i = a_i - b_{i-1}^2/L_{i-1}$ and $U_{N_g} = a_{N_g}$, $U_i = a_i - b_i^2/U_{i+1}$. Then $[A^{-1}]_{ii} = 1/(L_i + U_i - a_i)$.*

This is the standard recursion for the diagonal of a tridiagonal inverse [41]. It is also used in the recursive Green's function method [42], and in 1D it is equivalent to selected inversion [33].

In more than one dimension we rely on nearsightedness. In a gapped system the resolvent and the density matrix decay exponentially with distance [9, 10, 43], so the density in a region can be obtained from a local window [Figure 1(f)]. We do not prove a bound for truncated windows. We measure their error (Section VI).

III. ALGORITHMS

The self-consistent field (SCF) cycle is the same as in KS DFT, except for the step that turns a potential into a density (Figure 2). We describe this step first in one dimension, where g is computed exactly by a recursion, and then in three dimensions, where the density comes from eigenpairs of small windows.

A. One dimension

In 1D, $g(x, z_j)$ is computed at all grid points with Lemma 3 for $M/2$ nodes, and the density follows from Lemma 1; no eigenvectors are formed. The chemical potential is found by a secant search on $N(\mu)$, started from the previous SCF step. $E_{\rm lo}$ is the lowest eigenvalue, obtained by bisection, minus a margin δ.

B. Three dimensions: local windows

The 3D algorithm has four ingredients: a localized basis, a division of the system into regions with windows, the density from window eigenpairs weighted by r_M, and a chemical potential obtained from the electron count in closed form. We describe them in turn.

Basis. We use localized basis functions χ_μ tabulated on the grid, a linear combination of atomic orbitals (LCAO), as production linear-scaling codes do [16, 44]. The matrices are $H = X^{\mathsf{T}}(T_{\rm FD} + v)X\,\Delta V$ and $S = X^{\mathsf{T}}X\,\Delta V$. Here X holds the basis functions on the grid and $T_{\rm FD}$ is the FD kinetic operator.

Regions and windows. The system is divided into regions, one per molecule. Each grid point is assigned to the nearest molecule, that is, to the region B whose Voronoi cell contains it; we say that B owns these points. The density at an owned point is computed from the window $W(B)$, which contains the basis functions of all molecules within a radius R_w of the center of B.

Density from window eigenpairs. On each window we solve $H_W\phi_k = \lambda_k S_W\phi_k$ and set, for $\boldsymbol{r} \in B$,

$$n(\boldsymbol{r}) = 2\sum_k r_M\Big(\frac{\lambda_k - c}{R}\Big)|\phi_k(\boldsymbol{r})|^2. \tag{8}$$

By Lemma 1 this equals the pole sum $2\sum_j w_j g_W(\boldsymbol{r}, z_j)$ of the window Hamiltonian, so Equation (8) is a DC construction [13] with the occupation r_M. In 3D we therefore never form the local resolvent. Eigenpairs of a small window weighted by r_M are equivalent to summing M window resolvents and are about 20 times cheaper (Section VI). Here M enters only through the occupation function.

Electron number. The electron number and band energy, $N(\mu) = 2\sum_{B,k} c_{Bk} r_M(u_{Bk})$ and $E_{\rm band} = 2\sum_{B,k} c_{Bk}\lambda_{Bk} r_M(u_{Bk})$ with $c_{Bk} = \sum_{\boldsymbol{r}\in B}|\phi_k(\boldsymbol{r})|^2\Delta V$, are explicit functions of μ. The chemical potential is therefore found by solving $N(\mu) = N$ without further diagonalizations.

SCF steps. Once μ is known, only eigenpairs below $c + 1.5R$ are computed, because $r_M < 1.5^{-M}$ above that energy. Each SCF step then proceeds as follows.

1. In the first ("cold") step all window eigenvalues are computed, to place μ and $E_{\rm lo}$.

2. In later ("warm") steps μ and $E_{\rm lo}$ are reused, so only the low eigenpairs are computed.

3. If the electron count must be corrected, μ is resolved from the closed-form $N(\mu)$, and the density is rebuilt from the stored $|\phi_k(\boldsymbol{r})|^2$ on the owned points. No second eigensolver pass is needed.

4. E_{lo} is first set to the lowest window eigenvalue minus δ. It then stays fixed while the lowest window eigenvalue remains between $E_{\mathrm{lo}}+\delta/3$ and $E_{\mathrm{lo}}+2\delta$, that is, while the margin stays within a factor of 3 of δ. Otherwise the circle is re-centered: E_{lo} is reset to that eigenvalue minus δ.

Why not point by point? We first evaluated g point by point on a real-space grid (spacing 0.6 bohr) with probing vectors and Lanczos quadrature [19]. The quadrature itself was accurate, but probing needs far too many vectors: in our test crystal the density matrix extends over about 10^3 grid points, and with 125 probes (spacing 3 bohr) the error of the probed diagonal was 3.7 times the density itself, and with 1000 probes it was still 7% (Supplemental Material [38]). This is why we use a localized basis and windows.

Lemma 4 (Window spectra are bounded below)**.** *For a symmetric pencil (H, S) with $S > 0$, the lowest eigenvalue of any principal sub-pencil (H_W, S_W) is not lower than the lowest eigenvalue of (H, S).*

The lemma follows from the Courant–Fischer characterization (Appendix A). Truncating the Hamiltonian to a window therefore cannot create spurious states below the true band bottom. The choice of E_{lo} above places the circle below every window spectrum by construction, and within δ of the true band bottom as R_w grows.

IV. COMPUTATIONAL DETAILS

One dimension. Electrons interact through the soft-Coulomb potential $1/\sqrt{x^2+1}$ and are attracted by nuclei through $-Z/\sqrt{(x-X)^2+1}$, a common one-dimensional model [45]. The grid spacing is 0.1 bohr, with 6–10 bohr of vacuum and hard walls. We use the local exchange of the 1D soft-Coulomb uniform gas in the closed form $e_x(n) = -\pi^{-2}[bI_{K_0}(b) + bK_1(b) - 1]$, $b = \pi n$, $I_{K_0}(b) = \int_0^b K_0$ (Appendix B), without correlation. We refer to this as the exchange-only local-density approximation (LDA). Exact two-electron energies are obtained by diagonalization in the two-particle space on the same grid. The KS reference uses the identical Hamiltonian and is solved for the occupied states with LAPACK bisection (`stebz`) followed by inverse iteration (`stein`). This was faster than the multiple relatively robust representations (MRRR) algorithm (`stemr`) in our tests (Supplemental Material [38]). Unless stated otherwise $\delta = 0.3$ Ha.

Three dimensions.

System.: H_2 molecular crystals of $n \times n \times n$ cubic cells (lattice constant $a = 6$ bohr unless stated), with random molecular orientations and random displacements of 0.15 bohr, in periodic boundary conditions. The HOMO–LUMO gaps are 0.29–0.41 Ha.

Hamiltonian.: Nuclei are local pseudopotentials $-\mathrm{erf}(r/r_c)/r$ with $r_c = 0.6$ bohr. The grid spacing is 0.6 bohr, the Hartree potential is obtained by fast Fourier transform, and we use the LDA with Slater exchange and Perdew–Zunger correlation [46].

Basis.: Six functions per atom: s-type Gaussians with exponents 1.2, 0.4 and 0.15 bohr^{-2} and a p shell with exponent 0.5 bohr^{-2}, with a smooth cutoff at 4.5 bohr. Its error relative to a converged real-space-grid calculation, solved with the locally optimal block preconditioned conjugate gradient method (LOBPCG) [47], is 0.019 Ha per electron for 16 electrons.

KS reference.: KS and LRes-DFT use identical H, S, Hartree and xc terms. The dense KS reference solves the generalized eigenproblem for the lowest $N/2+1$ states. We compared three LAPACK paths: `dsygvx` (bisection and inverse iteration), `dsygvd` (divide and conquer) and Cholesky reduction with `dsyevr` (MRRR). We also compared two ways of building the density. We use the fastest combination, `dsygvx` with the orbital sum (Supplemental Material [38]). As a second reference we use LOBPCG on the pencil (H, S), warm-started from the previous SCF step, with a block-Jacobi preconditioner.

LRes-DFT.: $M = 64$, $\delta = 0.3$ Ha and $R_w = 12.1$ bohr unless stated.

Timing. Both 3D one-body solvers are written in C++ against the same OpenBLAS LAPACK, and the 1D LRes-DFT kernel is compiled with Numba. All timings are single-threaded unless stated, on an 8-core workstation. The headline comparison is the wall time of complete SCF calculations (Table 9). Each size was run three times, with the order of KS and LRes-DFT alternated, on an idle machine pinned to one core. The earlier density-build series of Table 11 and the thread timings were taken while the machine was shared (load average about 1); for those we report the best of two or three repetitions. In a separate series under heavier load, repeated KS timings varied by up to 20% (Supplemental Material [38]). Power-law exponents of the time are fitted to the four largest sizes, and those of the memory to the three largest. SCF calculations use Pulay mixing [48].

Verification. We checked the analytic identities with SymPy and 40-digit mpmath arithmetic on explicit instances (30 checks). The checks cover Lemmas 1–4, Propositions 1–3, the stationarity of G_M, the LDA potentials, the pseudopotential transform and the exchange closed form. Automated tests (25) check the implementation against these identities and against dense linear algebra.

V. RESULTS IN ONE DIMENSION

The 1D calculations test the pole rule itself. We check the error bound, count the complex solves needed by three pole rules, compare LRes-DFT with KS and with exact two-electron solutions, test forces, and measure the cost.

A. Error control

Figure 3(a) shows the density error of a 1D $(H_2)_{16}$ chain at its self-consistent KS potential as a function of M, together with the bound of Theorem 1. The error decreases exponentially, stays below the bound, and reaches 1.3×10^{-9} per electron at $M = 64$. The margin δ matters. With $M = 32$ the error per electron is 1.3×10^{-5} for $\delta = 0.3$ Ha and 2.1×10^{-7} for $\delta = 0.1$ Ha, which is about half the gap (Supplemental Material [38]). The results here use the conservative $\delta = 0.3$ Ha; choosing $\delta \approx \Delta/2$ would reduce the required M.

B. Comparison of pole rules

Table 2 and Figure 4 compare the number of complex solves needed for a density error of 10^{-6} per electron. The three rules are the half-step trapezoid rule ($M/2$ solves); Gauss–Legendre in the angle on the same circle (GL), the default of FEAST [35]; and Zolotarev's best rational approximation of the sign function (the occupation step is $[1 - \mathrm{sgn}(\varepsilon - \mu)]/2$, so a rational approximation of the sign gives a pole expansion) [36, 49] on $[-W, -\Delta/2] \cup [\Delta/2, W]$, centered at μ. All are evaluated exactly on the eigendecomposition of the same self-consistent KS Hamiltonian.

We consider three series. The first is grid refinement of an $(H_2)_4$ chain at fixed gap. The second is uniform H_n chains whose gap closes as n grows. The third is an $(H_2)_4$ chain with one heavier atom of charge Z, whose deep core state widens the occupied band; in this series the gap also decreases with Z.

Under grid refinement the half-step rule needs 18 solves at every spacing, while the spectral width grows from 50 to 3200 Ha. Gauss–Legendre on the same circle also does not depend on the grid and needs 13. Zolotarev's rule grows slowly (11 to 18), as expected from its $\ln(W/\Delta)$ dependence. As the gap closes, the half-step rule becomes much more expensive: 41 to 213 solves, against 20 to 38 for Gauss–Legendre and 15 to 16 for Zolotarev. Plotted against W_{occ}/Δ [Figure 4(c)], the H chains and the chains with a heavier atom fall on one curve, as Theorem 1 predicts. The cost of the half-step rule is therefore set by the ratio of occupied band width to gap, not by the deep state as such. The advantage of the half-step rule is that its occupations lie exactly in $[0, 1]$ (Lemma 2). At the accuracy considered, the Gauss–Legendre and Zolotarev occupations deviate from $[0, 1]$ by up to 2.4×10^{-5} and 1.7×10^{-6} , respectively (Supplemental Material [38]). We have not found a case in which these small deviations matter in practice. For small gaps or semicore states an optimal rational rule is the better choice. In the 3D window implementation, the rule affects only the occupation function and not the cost (Section III B).

Table 2. Complex solves for 10^{-6} density accuracy (1D) with the half-step rule (HS), Gauss–Legendre on the circle (GL) and Zolotarev's rule (Zol.). W: spectral width; Δ: gap; W_{occ}: occupied band width (Ha).

Case	W	Δ	W_{occ}	W_{occ}/Δ	HS	GL	Zol.
grid, $\Delta x = 0.2$	50	0.232	0.14	0.6	18	13	11
grid, $\Delta x = 0.1$	200	0.233	0.14	0.6	18	13	13
grid, $\Delta x = 0.05$	800	0.233	0.14	0.6	18	13	15
grid, $\Delta x = 0.025$	3200	0.233	0.14	0.6	18	13	18
H_{10} chain	200	0.095	0.24	2.5	41	20	15
H_{20} chain	200	0.052	0.27	5.3	68	25	15
H_{40} chain	200	0.027	0.29	10.6	119	33	16
H_{80} chain	200	0.014	0.30	21.2	213	38	16
core, $Z = 2$	200	0.229	0.17	0.7	19	13	13
core, $Z = 4$	200	0.107	0.91	8.5	75	26	14
core, $Z = 6$	200	0.053	1.86	34.8	247	48	15
core, $Z = 8$	200	0.045	2.88	64.5	413	63	15

Table 3. 1D two-electron systems: total energies (Ha) from exact diagonalization, KS and LRes-DFT ($M = 96$), with the same exchange-only LDA in KS and LRes-DFT.

System	Exact	KS	LRes	KS−ex.	|LRes−KS|
He	-2.23855	-2.15845	-2.15845	0.0801	3.6×10^{-15}
H_2, $R = 1.0$	-1.41320	-1.33295	-1.33295	0.0802	1.1×10^{-14}
H_2, $R = 1.6$	-1.45258	-1.37026	-1.37026	0.0823	1.2×10^{-14}
H_2, $R = 2.0$	-1.44200	-1.35599	-1.35599	0.0860	8.9×10^{-16}
H_2, $R = 3.0$	-1.38483	-1.27693	-1.27693	0.1079	9.8×10^{-15}
H_2, $R = 4.0$	-1.35067	-1.20590	-1.20590	0.1448	4.3×10^{-12}
H_2, $R = 6.0$	-1.33930	-1.13913	-1.13905	0.2002	7.8×10^{-5}

C. Accuracy

Table 3 and Figure 5(a) compare LRes-DFT and KS, with the same exchange-only LDA, with the exact solution of the interacting two-electron problem. LRes-DFT and KS agree to 5×10^{-12} Ha or better, except for stretched H_2 at $R = 6$ bohr, where the gap nearly closes and the difference is 8×10^{-5} Ha at $M = 96$. Both differ from the exact energies by 0.08–0.20 Ha, which is the error of the exchange functional. For LiH, Be and dimerized $(H_2)_n$ chains, self-consistent LRes-DFT energies agree with KS to 6×10^{-11} Ha per electron or better (Table 4). For a uniform H_{20} chain with a gap of 0.05 Ha the agreement is 8×10^{-8} Ha per electron at $M = 128$.

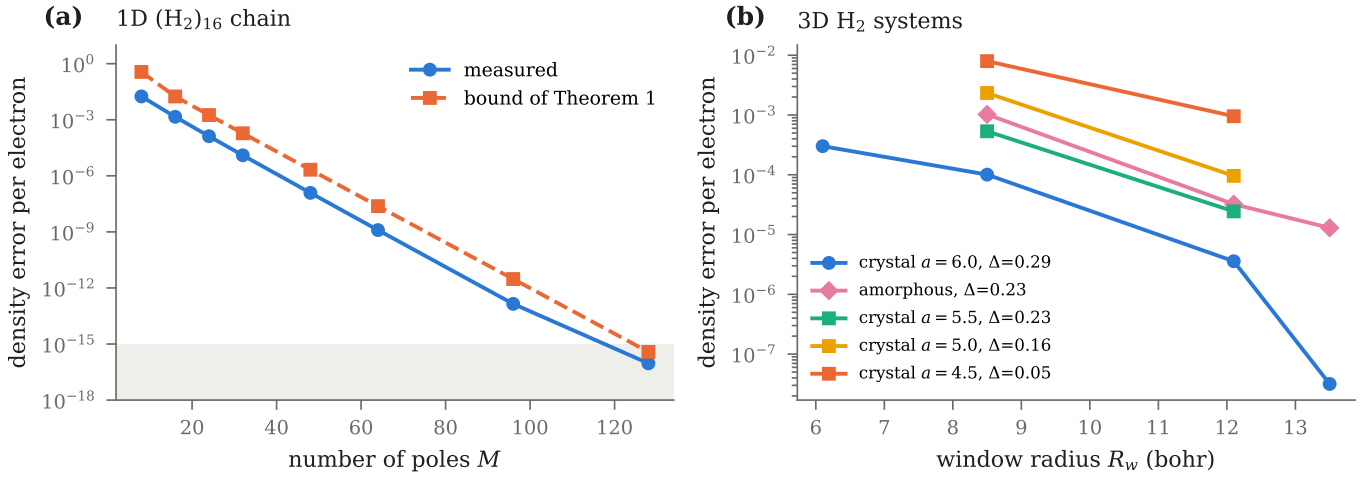


Figure 3. Error control. (a) 1D $(H_2)_{16}$ chain at the self-consistent KS potential ($\delta = 0.3$ Ha): density error per electron (circles) versus the number of poles M, and the bound of Theorem 1 (squares). (b) 3D H_2 systems with 250–432 electrons at their self-consistent KS potentials: density error per electron versus the window radius R_w for the crystal ($a = 6$ bohr), an amorphous arrangement at the same density, and compressed crystals ($a = 5.5$, 5.0 and 4.5 bohr). Gaps are given in the legend.

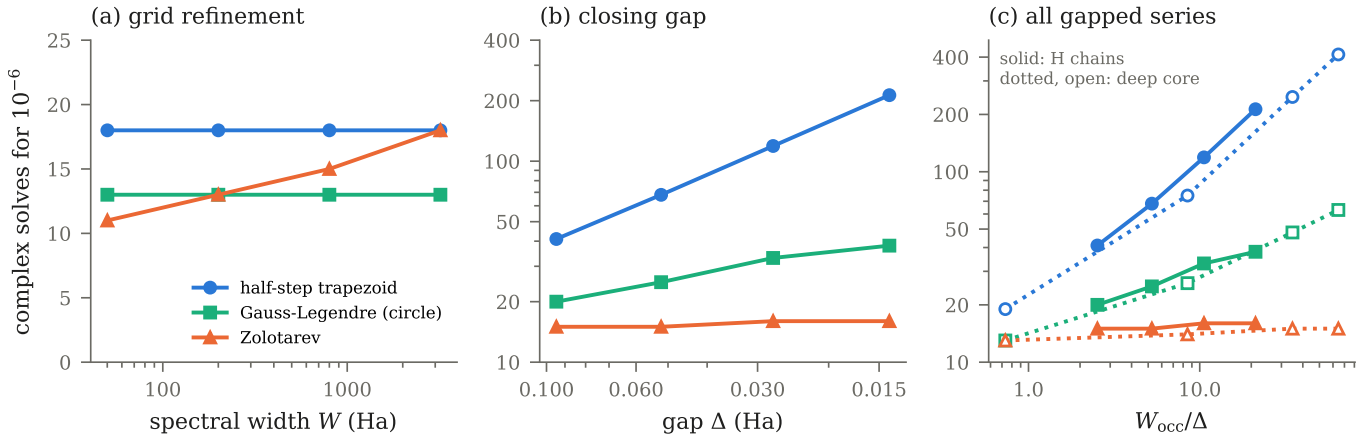


Figure 4. Number of complex solves needed for a density error of 10^{-6} per electron with the half-step trapezoid rule, Gauss–Legendre on the same circle, and Zolotarev's rational approximation. (a) Grid refinement at fixed gap (spectral widths 50–3200 Ha). (b) Closing gap in uniform H chains. (c) All gapped cases against the ratio W_{occ}/Δ of occupied band width to gap; solid lines and filled symbols: H chains; dotted lines and open symbols: chains with a heavier atom.

Table 4. 1D systems: self-consistent LRes-DFT ($M = 128$) versus KS on the identical Hamiltonian.

System	N	E_{KS} (Ha)	$\lvert\Delta E\rvert/N$ (Ha)	$\lVert\Delta n\rVert_1/N$
LiH	4	-4.836073	1.1×10^{-12}	5.0×10^{-12}
Be	4	-6.692767	5.2×10^{-11}	2.5×10^{-10}
$(H_2)_8$ chain	16	-10.983588	2.9×10^{-15}	5.1×10^{-15}
$(H_2)_{32}$ chain	64	-43.944324	8.9×10^{-16}	4.9×10^{-15}
H_{20} chain (small gap)	20	-13.818405	7.7×10^{-8}	9.1×10^{-7}

D. Forces

Table 5 tests Propositions 2 and 3 for a 1D H_2 molecule. Finite differences with steps of 10^{-2} and 10^{-3} bohr were extrapolated to zero step. At fixed μ and E_{lo}, the derivative of G_M equals the Hellmann–Feynman force to 10^{-10} Ha/bohr for $M = 8$, 16 and 64. At fixed N with a rigidly moving circle ($R = 0.5$ Ha), the derivative of A_M also equals the Hellmann–Feynman force to 10^{-10} Ha/bohr at every M, as Proposition 3 states. With E_{lo} fixed instead, the derivative of the fixed-N energy, Equation (2), differs from the Hellmann–Feynman force by 2.8×10^{-3}, 1.4×10^{-5} and 2×10^{-12} Ha/bohr for $M = 8$, 16 and 64. A rigidly moving circle should there-

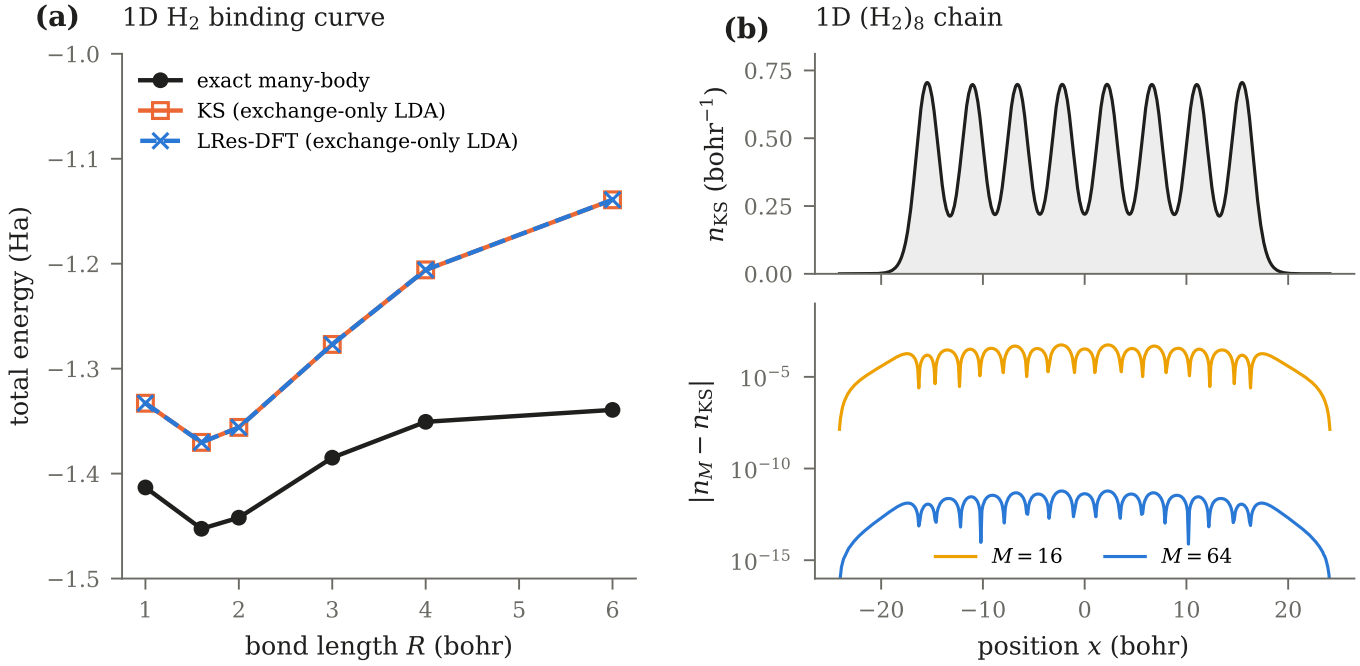


Figure 5. (a) Binding curve of 1D H_2: exact (black), KS (orange squares) and LRes-DFT (blue crosses), with the same exchange-only LDA in KS and LRes-DFT. (b) Self-consistent KS density of a 1D $(H_2)_8$ chain (top) and the LRes-DFT density error for $M = 16$ and 64 (bottom, logarithmic scale).

Table 5. Force on a nucleus of 1D H_2: difference (Ha/bohr) between the Hellmann–Feynman force and the extrapolated finite-difference derivative, at fixed μ (derivative of G_M), at fixed N with a rigidly moving circle (derivative of A_M), and at fixed N with fixed E_{lo} (derivative of Equation (2)).

M	fixed μ	fixed N, rigid circle	fixed N, fixed E_{lo}
8	2.9×10^{-11}	6.1×10^{-11}	2.8×10^{-3}
16	1.1×10^{-10}	2.5×10^{-11}	1.4×10^{-5}
64	8.6×10^{-11}	8.7×10^{-11}	2.2×10^{-12}

fore be used when forces are needed at small M. Forces in 3D, where the atom-centered basis adds basis-set (Pulay) force terms, were not computed.

E. Cost

LRes-DFT time grows as $N^{1.0}$–$N^{1.2}$ up to 8192 electrons [Figure 6(a)]. In 1D, LRes-DFT with Lemma 3 is a pole expansion with selected inversion, which is known to scale linearly in one dimension [33]. The comparison with KS eigensolvers therefore mainly confirms that the implementation reaches this scaling. The KS solver (bisection with inverse iteration) grows as $N^{2.8}$ here, because the occupied states of a dimerized chain form a narrow cluster. Our LRes-DFT timings include a secant search for μ that costs 25–47 density evaluations when started without a previous value. The 1D window variant of LRes-DFT also scales linearly, with an error of $(1.3\text{–}1.6) \times 10^{-7}$ per electron from 64 to 4096 electrons.

VI. RESULTS IN THREE DIMENSIONS

This section tests the window construction. We first measure the density error of a single window calculation at the self-consistent KS potential, for crystals, disordered systems and compressed systems. We then compare the windows with divide-and-conquer using Fermi–Dirac occupations, run complete SCF calculations, and measure the cost.

A. Window error, disorder and compression

Figure 3(b) and Table 6 show the density error per electron as a function of the window radius at the self-consistent KS potential. For the crystal at $a = 6$ bohr the error is 3.0×10^{-4}, 1.0×10^{-4}, 3.6×10^{-6} and 3.2×10^{-8} for $R_w = 6.1$, 8.5, 12.1 and 13.5 bohr. The decrease is stepwise because the molecules sit on a lattice, and the error is controlled by the nearest molecules outside the window.

In an amorphous arrangement at the same density, with windows centered on each molecule, the error at $R_w = 12.1$ bohr is 3.2×10^{-5}, nine times larger, and it decreases slowly (1.3×10^{-5} at 13.5 bohr). The self-

Table 6. 3D window error at the self-consistent KS potential ($M = 64$) for crystals with lattice constant a (bohr) and an amorphous arrangement. Δ: gap (Ha); n_W: basis functions per window; ΔE_b: band-energy difference (Ha).

System	N	Δ	R_w	n_W	$\|\Delta n\|_1/N$	$\|\Delta E_\mathrm{b}\|/N$
$a = 6.0$	250	0.29	8.5	228	1.0×10^{-4}	2.0×10^{-6}
$a = 6.0$	250	0.29	12.1	396	3.6×10^{-6}	1.5×10^{-7}
$a = 5.5$	432	0.23	8.5	228	5.3×10^{-4}	4.7×10^{-5}
$a = 5.5$	432	0.23	12.1	396	2.4×10^{-5}	1.6×10^{-7}
$a = 5.0$	432	0.16	8.5	228	2.3×10^{-3}	2.2×10^{-4}
$a = 5.0$	432	0.16	12.1	684	9.6×10^{-5}	4.6×10^{-6}
$a = 4.5$	432	0.05	8.5	324	8.0×10^{-3}	5.5×10^{-4}
$a = 4.5$	432	0.05	12.1	972	9.5×10^{-4}	1.5×10^{-5}
amorph.	250	0.23	8.5	146	1.0×10^{-3}	6.7×10^{-5}
amorph.	250	0.23	12.1	414	3.2×10^{-5}	2.8×10^{-7}

Table 7. Basis sets for the 3D crystal: functions per atom, cutoff (bohr), basis error (Ha/electron) relative to a converged grid calculation (16 electrons) and window error at $R_w = 12.1$ bohr. Exponents (bohr^{-2}): compact, s 1.0, 0.25 and p 0.6; default, s 1.2, 0.4, 0.15 and p 0.5; diffuse, s 1.0, 0.3, 0.12 and p 0.5.

Basis	Funcs./atom	Cutoff	Basis err.	Window err.
compact	5	4.5	0.051	1.3×10^{-7}
default	6	4.5	0.019	3.6×10^{-6}
diffuse	6	5.5	0.005	3.8×10^{-5}

consistent energy agrees with KS to 2.7×10^{-7} Ha per electron, with 20 SCF iterations for each method (convergence thresholds 10^{-8} for KS and 10^{-7} for LRes-DFT).

Compression is more severe. At $a = 5.5$, 5.0 and 4.5 bohr the gap falls to 0.23, 0.16 and 0.05 Ha. The error at $R_w = 12.1$ bohr then rises to 2.4×10^{-5}, 9.6×10^{-5} and 9.5×10^{-4}, while the window grows to 400–970 basis functions. Increasing M from 32 to 128 does not reduce the window error; it changes by at most a factor of 1.5 and increases slightly with M. The window, not the occupation function, controls the accuracy in 3D.

The window error also depends on the basis (Table 7 and Supplemental Material [38]). Compact and diffuse bases pull in opposite directions: a more compact basis has a larger basis error but a smaller window error, and a more diffuse basis the reverse. Diffuse functions make the overlap matrix nearly singular (smallest eigenvalue 0.004) and lengthen the range of the density matrix in the basis. In all cases studied the window spectra contain no states inside the KS gap, and their lowest eigenvalue lies above the global one, as Lemma 4 requires (Supplemental Material [38]).

Table 8. Density error per electron, 3D crystal (250 electrons), for the same windows with the occupation r_M ($M = 64$) and with Fermi–Dirac occupations (DC).

R_w (bohr)	n_W	r_M, $M = 64$	DC, $k_BT = 0.005$	DC, $k_BT = 0.01$
6.1	84	3.0×10^{-4}	3.0×10^{-4}	2.9×10^{-4}
8.5	228	1.0×10^{-4}	1.0×10^{-4}	1.0×10^{-4}
12.1	396	3.6×10^{-6}	3.4×10^{-6}	3.2×10^{-6}
13.5	684	3.2×10^{-8}	3.0×10^{-8}	6.0×10^{-8}

B. Relation to divide-and-conquer

Because Equation (8) is a DC construction, we compared it with DC using Fermi–Dirac occupations ($k_BT = 0.005$ and 0.01 Ha) in the same windows and at the same potential (Table 8). With $k_BT = 0.005$ Ha the density errors agree with those obtained with r_M to within 6% at every window radius. With $k_BT = 0.01$ Ha they agree to within 10%, except at the largest radius, where the smearing error of the Fermi–Dirac function exceeds the window error. Both occupation functions are evaluated on the same window eigenpairs, so their cost is the same. In 3D the occupation r_M offers no accuracy advantage over Fermi–Dirac occupations. The speed-up over KS reported below is a property of nearsighted DC methods in general.

C. Self-consistent calculations

Accuracy. Table 10 lists self-consistent results from 16 to 1024 electrons. For 16 and 54 electrons the window covers the whole periodic cell, so these rows test only the implementation. For 128 to 1024 electrons, LRes-DFT energies agree with KS to between 1.6×10^{-7} and 2.4×10^{-7} Ha per electron. The density difference, about 4.5×10^{-6} per electron, does not grow with system size, and both methods need the same number of SCF iterations (11–13). At 432 electrons the self-consistent energy error is 3.7×10^{-6}, 1.6×10^{-7} and 8.1×10^{-9} Ha per electron for $R_w = 8.5$, 12.1 and 13.5 bohr, so the window radius controls the self-consistent error as it does the non-self-consistent one.

Scaling and break-even. Table 9 gives the wall time of complete SCF calculations. The time scales as $N^{2.92\pm0.02}$ for dense KS and $N^{1.15\pm0.06}$ for LRes-DFT . The fits use the four largest KS sizes and the five largest LRes-DFT sizes, including one LRes-DFT run at 2000 electrons; the quoted error is the least-squares error of the slope only. Dense KS is faster below about 500 electrons, by a factor of 3 to 6 at 54–250 electrons. LRes-DFT is 2.0 times faster at 686 electrons and 3.5 times faster at 1024. Extrapolating the KS fit to 2000 electrons gives about 3×10^3 s, against 272 s measured for LRes-DFT. We did not run KS at that size, so this extrapolation should not

be quoted as a result.

Reliability of the timings. The three repeats of each size differ by less than 7%. Single timings taken earlier on a shared machine fluctuated by up to 1.5× (44 and 60 s for LRes-DFT at 432 electrons) and cannot be compared with these. The reference is single-threaded dense LAPACK (`dsygvx`). We did not time ELPA, a sparse solver or any other linear-scaling code, and a faster dense solver or a better-preconditioned iterative solver would move the break-even point to larger N. With eight threads, the KS step at 432 electrons went from 2.9 to 2.3 s and the LRes-DFT step from 2.7 to 1.0 s; these threaded timings come from a shared machine.

Accuracy against cost. The radius used for the headline numbers, $R_w = 12.1$ bohr, leaves an energy error of about 2×10^{-7} Ha per electron. A smaller error needs a larger window and more time: at 432 electrons, an energy error of 10^{-8} Ha per electron takes 159 s at $R_w = 13.5$ bohr against 44 s at 12.1 bohr and 39 s for KS (a separate run on a loaded machine), i.e. it is slower than dense KS .

Scope of the timings. All timings are for a compact Gaussian basis (six functions per electron) with local pseudopotentials, and $M = 64$ poles. Fewer poles would reduce the cost in proportion and increase the error (Section V B). We did not time nonlocal pseudopotentials, larger basis sets or diffuse functions, which enlarge the windows. The accompanying code also treats small molecules with GTH pseudopotentials (up to 10 atoms, 8 electrons). There the window covers the whole molecule and KS and LRes-DFT take the same time, about 7 s for H_2O at a grid spacing of 0.2 bohr, which is dominated by grid operations common to both. No speed-up is expected for such molecules.

Iterative reference. The warm-started iterative KS solver was slower than dense diagonalization in this basis (128 electrons: 3 s versus 2 s; 432 electrons: 44 s versus 34 s; at 1024 electrons a warm LOBPCG step took 46 s (up to 63 iterations) ; Supplemental Material [38]). The occupied states make up 1/12 of the basis, and the Rayleigh–Ritz step of LOBPCG on a block of about $3N/2$ vectors costs nearly as much as a dense diagonalization. Other iterative solvers or optimized dense libraries such as ELPA [50] may perform differently; we have not tested them.

D. Cost of a density build, memory and threads

Figure 6(b) and Table 11 give the cost of one density build. The fitted exponents are $N^{2.92}$ for KS (250–1024 electrons), $N^{1.05}$ for the LR-DFT warm step and $N^{1.02}$ for the cold step (432–2000 electrons) . The KS time is dominated by the reduction of the dense matrices to tridiagonal form, which costs $\mathcal{O}(N_b^3)$ for all three LAPACK paths tested. Building the density takes at most 8% of the KS time (Supplemental Material [38]).

The LRes-DFT prefactor is large, because every molecule lies in about 33 overlapping windows of about 400 basis functions each. Within a window, the eigendecomposition is much cheaper than the equivalent resolvent solves: for 396 basis functions it took 0.024 s, against 0.46–0.59 s for 32 complex solves, and the ratio was similar at 228 and 684 basis functions. There are three break-even points. For a single density build the crossover is at about 411 electrons for the warm step and about 685 electrons for the cold step; at 1024 electrons LR-DFT is 5.9 (warm) and 2.3 (cold) times faster . A complete SCF calculation consists of one cold step and several warm steps, so its break-even lies between them, near 500 electrons (Table 9). This last figure is the one we quote as the headline.

Table 9. Wall time (s, median and range of three runs) of a complete SCF calculation, single thread, idle machine, 11 SCF iterations for both methods; the last column is the ratio of the medians. The 2000-electron LRes-DFT value is a single run; dense KS was not run at that size.

N	KS	LRes-DFT	KS/LRes
54	0.5 (0.5–0.6)	3.2 (3.2–3.3)	0.17
128	1.9 (1.9–2.0)	8.6 (8.4–8.9)	0.22
250	7.2 (7.1–7.4)	25.0 (24.7–25.2)	0.29
432	34.6 (33.9–35.2)	46.0 (44.6–46.5)	0.75
686	132.0 (131.4–136.5)	65.8 (65.4–66.4)	2.01
1024	442.9 (441.2–445.1)	125.4 (125.0–126.0)	3.53
2000	–	272.1 (1 run)	–

The additional peak memory of the solver step grows as $N^{2.1}$ for KS and $N^{1.3}$ for LR-DFT; at 1024 electrons the solver step needs 676 MB (KS) and 251 MB (LR-DFT) in addition to the shared basis and grid data (Supplemental Material [38]). For 250 electrons and below the two methods use comparable memory. The windows are independent, and with OpenMP over windows the LRes-DFT step at 432 electrons went from 2.66 to 0.97 s on 8 threads (2.7×), while the threaded LAPACK of the KS step went from 2.90 s to 2.26 s at best (Supplemental Material [38]). These thread timings are rough, because the machine was shared and the threaded OpenBLAS is not the best-performing dense solver library.

VII. RELATION TO OTHER METHODS

Table 12 places LRes-DFT among existing approaches. Only the comparisons with KS (dense and LOBPCG), with DC using Fermi–Dirac occupations, and with the Gauss–Legendre and Zolotarev rules were made in this work.

The formulation is closest to the Green's-function methods. KKR-based methods such as LSMS and LSGF [24–26] also compute site-diagonal Green's functions on a complex contour in local zones, and they obtain the band energy from a contour integral. They use a

Table 10. Self-consistent 3D results: LRes-DFT ($M = 64$, $R_w = 12.1$ bohr) versus dense KS in the identical basis. Wall times are for the complete SCF calculation (single thread), from one run each on a shared machine. Use Table 9, which has repeated timings, for speed-ups.

N	E_{KS} (Ha)	$\lvert\Delta E\rvert/N$ (Ha)	$\lVert\Delta n\rVert_1/N$	SCF it. KS/LRes	KS wall (s)	LRes wall (s)	KS/LRes
16	-4.634688	6.5×10^{-10}	9.7×10^{-9}	13/17	0.2	0.4	0.51
54	-15.746533	2.1×10^{-13}	8.7×10^{-9}	13/16	0.7	5.6	0.13
128	-37.357441	2.4×10^{-7}	4.8×10^{-6}	13/13	2.5	14.8	0.17
432	-126.019962	1.6×10^{-7}	4.6×10^{-6}	11/11	39	44	0.89
686	-200.257102	1.9×10^{-7}	4.4×10^{-6}	11/11	136	62	2.20

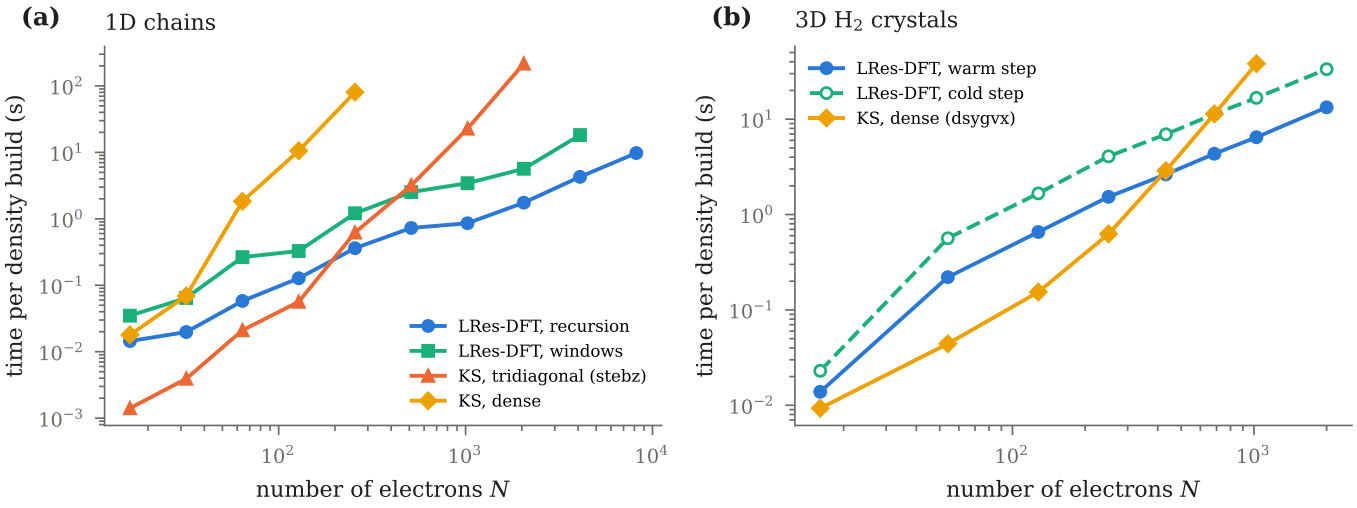


Figure 6. Time for one density build (single thread). (a) 1D dimerized chains: LRes-DFT with the exact recursion and with local windows, and KS with a tridiagonal eigensolver and with dense diagonalization. (b) 3D H_2 crystals in the identical basis: LRes-DFT warm and cold steps and dense KS (`dsygvx`).

Table 11. 3D cost of one density build (s, single thread, best of two or three) and the LRes-DFT–KS density difference per electron at the initial-guess potential. N_b: basis functions.

N	N_b	KS	LRes warm	LRes cold	$\lVert\Delta n\rVert_1/N$
16	96	0.01	0.01	0.02	3.3×10^{-15}
54	324	0.04	0.22	0.56	1.7×10^{-15}
128	768	0.15	0.66	1.67	8.0×10^{-7}
250	1500	0.63	1.53	4.07	6.3×10^{-7}
432	2592	2.88	2.64	6.94	7.2×10^{-7}
686	4116	11.35	4.34	11.33	7.9×10^{-7}
1024	6144	38.20	6.46	16.73	7.8×10^{-7}
2000	12000	–	13.30	33.41	–

multiple-scattering representation, whereas here the resolvent is taken in a grid or localized basis. Pole expansions with selected inversion [29, 33] evaluate the same contour integral with optimal poles and no locality truncation. Our 1D implementation is of this type with a non-optimal rule, and optimal rules need fewer poles (Section V B). In 3D our implementation is a DC method [13], equivalent in accuracy and cost to DC with Fermi–Dirac occupations (Section VI B). Fragment methods such as LS3DF [14] use a patching scheme that cancels boundary errors, which we have not tested. Compared with orbital-free DFT, LRes-DFT keeps the exact KS kinetic energy, at a much larger cost per electron.

VIII. LIMITATIONS

Small gaps and metals.: The number of poles of the half-step rule grows as W_{occ}/Δ, and the window error grows as the gap closes. At a gap of 0.05 Ha (compressed H_2) the 3D density error is 9.5×10^{-4} per electron at $R_w = 12.1$ bohr. Metals, and with them the alloy problems that motivate this work, were not studied. They would need finite electronic temperature and an optimal pole rule.

Disorder and basis set.: The window error is nine times larger for an amorphous arrangement than for the crystal, and it grows with the diffuseness of the basis.

Table 12. LRes-DFT and related approaches. "Extra error" is the error beyond that of the xc functional and basis. Entries other than KS, DC and LRes-DFT summarize the cited literature and were not measured here.

Method	Variable	Kinetic energy	Cost (3D, gapped)	Extra error and its control
KS, dense diagonalization	$N/2$ orbitals	exact	$\mathcal{O}(N^3)$	none (measured here: $N^{2.9}$)
Orbital-free DFT [6–8]	$n(\boldsymbol{r})$	approximate	$\mathcal{O}(N \ln N)$	kinetic functional
Density matrix / localized orbitals [15–18]	sparse density matrix or orbitals	exact	$\mathcal{O}(N)$	localization radius
Divide-and-conquer, fragments [13, 14]	local densities	exact	$\mathcal{O}(N)$	buffer radius (DC measured here)
Chebyshev / Fermi operator [11, 19]	polynomial of $\hat{h}$	exact	$\mathcal{O}(N)$	degree $\propto$ spectral width
Pole expansion + selected inversion [29, 31, 33]	resolvents	exact	$\mathcal{O}(N^2)$	pole number $\propto \ln(W/\Delta)$
KKR-based (LSMS, LSGF) [24–26]	site Green's functions	exact	$\mathcal{O}(N)$	local-zone size
Stochastic DFT [22]	random traces	exact	$\lesssim \mathcal{O}(N)$	statistical noise
LRes-DFT (this work)	local resolvent g	exact	$\mathcal{O}(N)$	ρ^M (rule) and window radius

Scope.: Only local pseudopotentials and the LDA were used. Nonlocal operators need off-diagonal density-matrix elements (Remark 2). The 3D tests use only H_2 molecular solids. The basis has its own error of 0.019 Ha per electron.

Forces.: Forces were verified in 1D only. 3D forces, which need the basis-set (Pulay) terms, were not computed.

Prefactor and references.: In 3D, LRes-DFT is slower than dense KS below the break-even size, and its gain above it is modest (Table 9). The gain relative to a well-tuned dense or iterative solver, and for larger or nonlocal-pseudopotential bases, is unknown. We did not compare with ELPA [50], ELSI [51] or other linear-scaling codes, and parallel efficiency was not measured on a dedicated machine.

IX. CONCLUSIONS

We have written the Kohn–Sham density in terms of the diagonal of the one-electron resolvent and analyzed a half-step trapezoid rule for it. The density and band energy are linear in the local resolvent, the occupations lie in $[0, 1]$ for every M, and a circle that moves rigidly with the chemical potential gives exact fixed-N forces. In one dimension the error obeys an a priori bound that depends on the grid only logarithmically. Gauss–Legendre and Zolotarev rules need fewer nodes, and the cost of the half-step rule grows with the ratio of occupied band width to gap.

In three dimensions the method is a divide-and-conquer scheme with the same accuracy as divide-and-conquer with Fermi–Dirac occupations. For gapped H_2 molecular solids it reproduces self-consistent Kohn–Sham energies to 2.5×10^{-7} Ha per electron up to 1024 electrons. The cost grows linearly with system size, but the prefactor is large: on one core the method breaks even with dense diagonalization near 500 electrons and is 3.5 times faster at 1024 electrons. The window error grows with disorder, compression and basis diffuseness, and it is the main limitation in three dimensions. Extending the approach to metals will require finite electronic temperature and an optimal pole rule.

ACKNOWLEDGMENTS

The code, data, computer-algebra scripts and tests are available at https://github.com/albert-hzbn/LResDFT.

Appendix A: Proofs

Notation is that of Section II: $u_i = (\varepsilon_i - c)/R$, $r_M(u) = 1/(1+u^M)$, $\zeta_j = \mathrm{e}^{\mathrm{i}\pi(2j+1)/M}$ $(j = 0, \ldots, M-1)$, $z_j = c + R\zeta_j$ and $w_j = R\zeta_j/M$. Two elementary facts are used repeatedly.

(F1) $\zeta_j^M = \mathrm{e}^{\mathrm{i}\pi(2j+1)} = -1$, and the M numbers ζ_j are distinct, so they are *all* the roots of $\zeta^M + 1$, and each is simple.

(F2) $\sum_j \zeta_j = 0$, because $\zeta_j = \zeta_0\, \omega^j$ with $\omega = \mathrm{e}^{2\pi\mathrm{i}/M} \neq 1$, and $\sum_j \omega^j = (1-\omega^M)/(1-\omega) = 0$. Hence $\sum_j w_j = 0$.

Also, $\zeta_{M-1-j} = \overline{\zeta_j}$, so the nodes and weights come in complex-conjugate pairs.

1. Closed form and bounds (Lemmas 1 and 2)

Proof of Lemma 1. Step 1: residues. Write $f(u) = p(u)/q(u)$ with $p = 1$ and $q = 1 + u^M$. By (F1) the zeros of q are simple, so the residue at ζ_j is p/q':

$$\begin{aligned}\mathrm{Res}_{u=\zeta_j} f &= \frac{1}{M\zeta_j^{M-1}} = \frac{\zeta_j}{M\,\zeta_j^{M}} \\ &= -\frac{\zeta_j}{M}. \end{aligned} \tag{A1}$$

Step 2: partial fractions. Let $s(u) = \sum_j \mathrm{Res}_{\zeta_j} f/(u - \zeta_j)$. The difference $f - s$ has no poles, so it is entire; both f and s tend to 0 as $|u| \to \infty$, so $f - s$ is bounded and tends to 0. By Liouville's theorem, $f - s \equiv 0$:

$$f(u) = -\frac{1}{M}\sum_j \frac{\zeta_j}{u - \zeta_j} = \frac{1}{M}\sum_j \frac{\zeta_j}{\zeta_j - u}. \tag{A2}$$

Step 3: change of variable. Put $u = (\varepsilon - c)/R$. Multiplying numerator and denominator by R and using $R\zeta_j = z_j - c$,

$$\begin{aligned}\frac{1}{M}\frac{\zeta_j}{\zeta_j - u} &= \frac{R\zeta_j}{M}\,\frac{1}{R\zeta_j - (\varepsilon - c)} \\ &= \frac{w_j}{z_j - \varepsilon}. \end{aligned} \tag{A3}$$

Summing over j gives the scalar identity

$$r_M(u) = \sum_j \frac{w_j}{z_j - \varepsilon}. \tag{A4}$$

Step 4: matrix form. Let $\hat{h} = \sum_i \varepsilon_i |\phi_i\rangle\langle\phi_i|$. Then $(z_j - \hat{h})^{-1} = \sum_i |\phi_i\rangle\langle\phi_i|/(z_j - \varepsilon_i)$, and by Equation (A4)

$$\sum_j w_j (z_j - \hat{h})^{-1} = \sum_i r_M(u_i)\,|\phi_i\rangle\langle\phi_i| = r_M\Big(\frac{\hat{h} - c}{R}\Big), \tag{A5}$$

which is Equation (4).

Step 5: non-orthogonal basis. Since $S > 0$, the matrix $S^{-1}H = S^{-1/2}\big(S^{-1/2}HS^{-1/2}\big)S^{1/2}$ is similar to a real symmetric matrix, so $S^{-1}H = X\Lambda X^{-1}$ with real diagonal Λ. Then $(zS - H)^{-1} = (z - S^{-1}H)^{-1}S^{-1} = X(z - \Lambda)^{-1}X^{-1}S^{-1}$, and applying Equation (A4) to each diagonal entry of Λ gives $\sum_j w_j (z_j S - H)^{-1} = X\, r_M(\Lambda) X^{-1}S^{-1} = r_M(S^{-1}H)S^{-1}$. □

Proof of Lemma 2. Step 1: range. For real u and even M, $u^M = (u^2)^{M/2} = |u|^M \geq 0$. Therefore $1 + u^M \geq 1$ and $0 < r_M \leq 1$. The eigenvalues of $\hat{P}_M$ are the numbers $r_M(u_i) \in (0, 1]$, so $\hat{P}_M$ is Hermitian with $0 \preceq \hat{P}_M \preceq 1$. Together with the factor 2 for spin, this is the definition of an ensemble-representable density matrix.

Step 2: states inside the circle. If $|u| \leq \rho < 1$, then

$$1 - r_M = \frac{|u|^M}{1 + |u|^M} \leq |u|^M \leq \rho^M. \tag{A6}$$

Step 3: states outside the circle. If $|u| \geq 1/\rho > 1$, then

$$r_M = \frac{1}{1 + |u|^M} \leq |u|^{-M} \leq \rho^M. \tag{A7}$$

Example. For $M = 16$ and $u = 0.8$: $u^M = 0.0282$ and $r_M = 0.973$; for $u = 1.25$: $u^M = 35.5$ and $r_M = 0.027$. For $M = 64$ these become $1 - 6.3 \times 10^{-7}$ and 6.3×10^{-7}. □

2. Density and band energy (Proposition 1, Remark 2)

Proof of Proposition 1. Step 1: residues of g. By Equation (1), $g(\boldsymbol{r}, z) = \sum_i |\phi_i(\boldsymbol{r})|^2/(z - \varepsilon_i)$ has a simple pole at each ε_i with residue $|\phi_i(\boldsymbol{r})|^2$. The function $z\,g$ has the residue $\varepsilon_i|\phi_i(\boldsymbol{r})|^2$ there, because $z = \varepsilon_i$ at the pole.

Step 2: residue theorem. For a counterclockwise contour C that encloses the occupied levels and no others,

$$\frac{1}{2\pi\mathrm{i}}\oint_C g\,dz = \sum_{i\,\mathrm{occ}} |\phi_i(\boldsymbol{r})|^2, \tag{A8}$$

$$\frac{1}{2\pi\mathrm{i}}\oint_C z\,g\,dz = \sum_{i\,\mathrm{occ}} \varepsilon_i|\phi_i(\boldsymbol{r})|^2. \tag{A9}$$

Multiplying the first by 2 gives (a).

Step 3: sum over grid points. Because $\sum_{\boldsymbol{r}} |\phi_i(\boldsymbol{r})|^2 = 1$ for each normalized ϕ_i, summing the second line over $\boldsymbol{r}$ and multiplying by 2 gives (b).

Step 4: the discretized rule. Replacing $\frac{1}{2\pi\mathrm{i}}\oint_C(\cdot)\,dz$ by $\sum_j w_j(\cdot)|_{z_j}$ and using Lemma 1, the density is $n_M\Delta V = 2\sum_j w_j g(\boldsymbol{r}, z_j) = 2\sum_i r_M(u_i)|\phi_i(\boldsymbol{r})|^2$. For the band energy, write $\hat{h} = z_j - (z_j - \hat{h})$ inside the trace:

$$\begin{aligned}\mathrm{Tr}[\hat{h}\hat{P}_M] &= \sum_j w_j\,\mathrm{Tr}\left[\hat{h}\,(z_j - \hat{h})^{-1}\right] \\ &= \sum_j w_j\,\mathrm{Tr}\left[z_j(z_j - \hat{h})^{-1} - 1\right] \\ &= \sum_j w_j z_j\,\mathrm{Tr}(z_j - \hat{h})^{-1} - N_g\sum_j w_j. \end{aligned} \tag{A10}$$

By (F2) the last term vanishes. Since $\mathrm{Tr}(z - \hat{h})^{-1} = \sum_{\boldsymbol{r}} g(\boldsymbol{r}, z)$, the band energy $2\,\mathrm{Tr}[\hat{h}\hat{P}_M] = 2\sum_i \varepsilon_i r_M(u_i)$ is again linear in g. □

Proof of Remark 2. Step 1: Neumann series. For $|z| > \|\hat{h}\|$, $(z - \hat{h})^{-1} = z^{-1}(1 - \hat{h}/z)^{-1} = \sum_{k\geq 0}\hat{h}^k/z^{k+1}$, convergent in operator norm.

Step 2: expansion of $z\,g$. Taking the $\langle\boldsymbol{r}|\cdot|\boldsymbol{r}\rangle$ element and multiplying by z,

$$z\,g(\boldsymbol{r}, z) = 1 + \frac{h_{\boldsymbol{rr}}}{z} + \frac{(\hat{h}^2)_{\boldsymbol{rr}}}{z^2} + \cdots. \tag{A11}$$

Step 3: contour integral. On $|z| = R'$ with $R' > \|\hat{h}\|$, termwise integration is allowed. Only the $1/z$ term survives, since $\frac{1}{2\pi\mathrm{i}}\oint z^{-k}dz = \delta_{k1}$, and it gives $h_{\boldsymbol{rr}}$. Finally $h_{\boldsymbol{rr}} = t_{\boldsymbol{rr}} + v(\boldsymbol{r})$ for a local potential. □

3. Stationarity and forces (Propositions 2 and 3)

Proof of Proposition 2 and of the stationarity of G_M. *Step 1: properties of W_M.* Differentiating term by term,

$$W'_M(\varepsilon) = 2\sum_j \frac{w_j}{z_j - \varepsilon} = 2r_M(u), \tag{A12}$$

by Equation (A4). No node is real, so each logarithm is analytic along the real axis. The conjugate pairing of nodes and weights makes the sum real.

Step 2: derivative of a spectral sum. Let $\Omega_M = \sum_i W_M(\varepsilon_i)$. First-order perturbation theory gives $\partial\varepsilon_i/\partial v(\boldsymbol{r}) = \langle\phi_i|\boldsymbol{r}\rangle\langle\boldsymbol{r}|\phi_i\rangle = |\phi_i(\boldsymbol{r})|^2$, because $\partial\hat{h}/\partial v(\boldsymbol{r}) = |\boldsymbol{r}\rangle\langle\boldsymbol{r}|$. Hence

$$\begin{aligned}\frac{\partial\Omega_M}{\partial v(\boldsymbol{r})} &= \sum_i W'_M(\varepsilon_i)\,|\phi_i(\boldsymbol{r})|^2 \\ &= 2\sum_i r_M(u_i)|\phi_i(\boldsymbol{r})|^2 \\ &= n_M(\boldsymbol{r})\Delta V.\end{aligned} \tag{A13}$$

For a single pole, the same argument is Jacobi's formula $\partial \ln\det(z-\hat{h})/\partial v(\boldsymbol{r}) = -g(\boldsymbol{r},z)$.

Step 3: derivative of G_M. Define $\chi_{\boldsymbol{r}\boldsymbol{r}'} = \partial n_M(\boldsymbol{r})/\partial v(\boldsymbol{r}')$. In Equation (6), E_{H} and E_{xc} depend on v only through n_M, with $\partial(E_{\mathrm{H}} + E_{\mathrm{xc}})/\partial n(\boldsymbol{r}) = v_{\mathrm{Hxc}}(\boldsymbol{r})\Delta V$. The product rule on the middle term gives

$$\begin{aligned}\frac{\partial G_M}{\partial v(\boldsymbol{r}')} &= n_M(\boldsymbol{r}')\Delta V - n_M(\boldsymbol{r}')\Delta V \\ &\quad - \sum_{\boldsymbol{r}} (v - v_{\mathrm{ext}})(\boldsymbol{r})\,\chi_{\boldsymbol{r}\boldsymbol{r}'}\Delta V \\ &\quad + \sum_{\boldsymbol{r}} v_{\mathrm{Hxc}}[n_M](\boldsymbol{r})\,\chi_{\boldsymbol{r}\boldsymbol{r}'}\Delta V.\end{aligned} \tag{A14}$$

The first line uses Equation (A13) and cancels. Collecting the rest,

$$\frac{\partial G_M}{\partial v(\boldsymbol{r}')} = \sum_{\boldsymbol{r}} \chi_{\boldsymbol{r}\boldsymbol{r}'}\big[v_{\mathrm{ext}} + v_{\mathrm{Hxc}}[n_M] - v\big](\boldsymbol{r})\,\Delta V. \tag{A15}$$

The bracket vanishes at self-consistency, $v = v_{\mathrm{ext}} + v_{\mathrm{Hxc}}[n_M]$, so G_M is stationary. No assumption on χ is needed.

Step 4: force. Let X be a nuclear coordinate. At fixed v, Ω_M and n_M do not depend on X, and E_{H} and E_{xc} depend on X only through n_M. So X enters G_M only through v_{ext} in the middle term and through E_{nn}. By the chain rule and Step 3,

$$\begin{aligned}\frac{dG_M}{dX} &= \frac{\partial G_M}{\partial X}\Big|_v + \sum_{\boldsymbol{r}'} \underbrace{\frac{\partial G_M}{\partial v(\boldsymbol{r}')}}_{=0} \frac{dv(\boldsymbol{r}')}{dX} \\ &= \sum_{\boldsymbol{r}} n_M\,\frac{\partial v_{\mathrm{ext}}}{\partial X}\,\Delta V + \frac{\partial E_{nn}}{\partial X},\end{aligned} \tag{A16}$$

which is the Hellmann–Feynman expression. □

Proof of Proposition 3. Step 1: μ-dependence of the rule. With R fixed and $c = \mu - R$, the nodes are $z_j = \mu - R + R\zeta_j$ and the weights $w_j = R\zeta_j/M$ do not depend on μ. In Equation (A4) only $u = (\varepsilon - \mu)/R + 1$ depends on μ, so W_M is a function of $\varepsilon - \mu$ and

$$\frac{\partial W_M}{\partial\mu} = -W'_M = -2r_M. \tag{A17}$$

Step 2: electron number. Summing over the eigenvalues,

$$\frac{\partial\Omega_M}{\partial\mu} = -\sum_i 2r_M(u_i) = -N_M, \tag{A18}$$

$$\frac{\partial A_M}{\partial\mu} = -N_M + N, \tag{A19}$$

which vanishes when $N_M = N$.

Step 3: stationarity in v. At fixed μ the circle is fixed, so Step 3 of the previous proof applies unchanged, and μN does not depend on v.

Step 4: force. A_M is stationary with respect to v and μ at the self-consistent solution, so its total derivative with respect to X equals its explicit derivative, which is the Hellmann–Feynman expression of the previous proof. The force is $-dA_M/dX$, exactly at every M. □

4. Convergence (Theorem 1)

Proof of Theorem 1. Idea. The density error is bounded by the sum of the occupation errors of all eigenstates. Occupied states lie inside the circle and each contributes at most ρ^M. Empty states close to the circle also contribute at most ρ^M, and empty states far from it at most 2^{-M}.

Step 1: the circle in terms of the gap. From $E_{\mathrm{lo}} = c - R$ and $\mu = c + R$ we get $2R = \mu - E_{\mathrm{lo}} = \Delta_- + (\varepsilon_{\mathrm{HOMO}} - E_{\mathrm{min}}) + \delta$. The scaled energies of the three landmark states are

$$\begin{aligned}u_{\mathrm{min}} &= \frac{E_{\mathrm{min}} - c}{R} = -1 + \frac{\delta}{R}, \\ u_{\mathrm{HOMO}} &= \frac{\varepsilon_{\mathrm{HOMO}} - \mu}{R} + 1 = 1 - \frac{\Delta_-}{R}, \\ u_{\mathrm{LUMO}} &= \frac{\varepsilon_{\mathrm{LUMO}} - \mu}{R} + 1 = 1 + \frac{\Delta_+}{R}.\end{aligned} \tag{A20}$$

Step 2: reduction to eigenvalues. Let $\hat{P}$ be the exact projector on the occupied states and $A = \hat{P}_M - \hat{P}$. Then $(n_M - n)\Delta V = 2\,\mathrm{diag}\,A$. A has the eigenvectors ϕ_i of $\hat{h}$ and eigenvalues $a_i = r_M(u_i) - \theta_i$, with $\theta_i = 1$ for occupied and 0 for empty states. By the triangle inequality and

normalization,

$$\sum_{\boldsymbol{r}} |A_{\boldsymbol{r}\boldsymbol{r}}| = \sum_{\boldsymbol{r}} \Big| \sum_i a_i |\phi_i(\boldsymbol{r})|^2 \Big| \le \sum_i |a_i| \sum_{\boldsymbol{r}} |\phi_i(\boldsymbol{r})|^2 = \sum_i |a_i|. \tag{A21}$$

Step 3: occupied states. The map $\varepsilon \mapsto u$ is increasing, so occupied states lie in $[u_{\min}, u_{\mathrm{HOMO}}]$ and $|u_i| \le \max(|1-\delta/R|, 1-\Delta_-/R) \le \rho$. By Lemma 2, $|a_i| = 1 - r_M \le \rho^M$. There are $N_{\mathrm{occ}} = N/2$ such states.

Step 4: empty states. These satisfy $u_i \ge u_{\mathrm{LUMO}} = 1 + \Delta_+/R$, hence $|u_i|^{-1} \le 1/(1+\Delta_+/R) \le \rho$. In the first group, $1 < |u_i| < 2$ (N_{near} states), Lemma 2 gives $|a_i| = r_M \le \rho^M$. In the second, $|u_i| \ge 2$ (at most N_g states), $|a_i| = r_M \le |u_i|^{-M} \le 2^{-M}$.

Step 5: sum. Adding the three contributions and multiplying by 2,

$$\sum_{\boldsymbol{r}} |n_M - n| \Delta V \le 2\,(N_{\mathrm{occ}} + N_{\mathrm{near}})\,\rho^M + 2N_g\,2^{-M}. \tag{A22}$$

□

5. Recursion and window spectra (Lemmas 3 and 4)

Proof of Lemma 3. Step 1: pivots are inverse corner elements. Let $A_{[1:k]}$ be the leading $k \times k$ block of A. By Gaussian elimination from the top, the kth pivot is the Schur complement of $A_{[1:k-1]}$ in $A_{[1:k]}$. For a tridiagonal matrix only b_{k-1} couples row k to the block, so

$$L_k = a_k - b_{k-1}^2 \left[A_{[1:k-1]}^{-1}\right]_{k-1,k-1}. \tag{A23}$$

Applying the block-inverse formula to $A_{[1:k]}$ shows $\left[A_{[1:k]}^{-1}\right]_{kk} = 1/L_k$. Induction on k then gives the recursion $L_k = a_k - b_{k-1}^2/L_{k-1}$ with $[A_{[1:k-1]}^{-1}]_{k-1,k-1} = 1/L_{k-1}$. The same argument from the bottom gives $U_k = a_k - b_k^2/U_{k+1}$ and $[A_{[k+1:N_g]}^{-1}]_{k+1,k+1} = 1/U_{k+1}$.

Step 2: diagonal element. Partition A around row i into the top block, the ith row and the bottom block. The only couplings of row i are b_{i-1} to the last row of the top block and b_i to the first row of the bottom block. The (i,i) element of A^{-1} is the inverse of the Schur complement of the two outer blocks:

$$[A^{-1}]_{ii} = \Big[a_i - \frac{b_{i-1}^2}{L_{i-1}} - \frac{b_i^2}{U_{i+1}}\Big]^{-1} = \frac{1}{L_i + U_i - a_i}, \tag{A24}$$

where the last step substitutes $L_i - a_i = -b_{i-1}^2/L_{i-1}$ and $U_i - a_i = -b_i^2/U_{i+1}$. See also Ref. [41]. □

Proof of Lemma 4. Step 1: variational form. With $x = S^{-1/2}y$, the generalized problem becomes a standard one,

$$\lambda_{\min}(H,S) = \min_{x \ne 0} \frac{x^{\mathsf{T}} H x}{x^{\mathsf{T}} S x} = \lambda_{\min}\big(S^{-1/2} H S^{-1/2}\big), \tag{A25}$$

by the Rayleigh–Ritz (Courant–Fischer) principle. The same holds for (H_W, S_W).

Step 2: restriction. Let E be the matrix whose columns embed the window basis functions in the full basis, so that $H_W = E^{\mathsf{T}} H E$ and $S_W = E^{\mathsf{T}} S E$. For $y \ne 0$ in the window space, $x = Ey$ is a nonzero vector (columns of E are linearly independent), and

$$\frac{y^{\mathsf{T}} H_W y}{y^{\mathsf{T}} S_W y} = \frac{x^{\mathsf{T}} H x}{x^{\mathsf{T}} S x}. \tag{A26}$$

Step 3: conclusion. The window minimum is therefore taken over a subset of the vectors used for the full minimum, so

$$\lambda_{\min}(H_W, S_W) \ge \lambda_{\min}(H,S). \tag{A27}$$

□

Appendix B: Exchange energy of the 1D soft-Coulomb gas

We consider the uniform gas of density n (spin-unpolarized) with interaction $w(r) = 1/\sqrt{r^2+1}$.

Step 1: density matrix. The occupied plane waves fill $|k| \le k_F$, and the spin-summed density is $n = 2 \cdot 2k_F/(2\pi)$, so $k_F = \pi n/2$. The spin-summed one-body density matrix is

$$\gamma(r) = 2\int_{-k_F}^{k_F} \frac{dk}{2\pi}\, \mathrm{e}^{\mathrm{i}kr} = \frac{2\sin(k_F r)}{\pi r}. \tag{B1}$$

Step 2: exchange energy as an integral. The exchange energy per unit length is $e_x = -\frac{1}{4}\int_{-\infty}^{\infty} w(r)\,\gamma(r)^2\,dr$. Using $\gamma^2 = 4\sin^2(k_F r)/(\pi^2 r^2)$ and the evenness of the integrand,

$$e_x = -\frac{2}{\pi^2}\int_0^\infty \frac{\sin^2(k_F r)}{r^2\sqrt{1+r^2}}\,dr = -\frac{1}{\pi^2}\,I(2k_F), \tag{B2}$$

where we used $\sin^2(k_F r) = \frac{1}{2}[1 - \cos(2k_F r)]$ and defined

$$I(b) = \int_0^\infty \frac{1-\cos br}{r^2\sqrt{1+r^2}}\,dr. \tag{B3}$$

Step 3: differential equation for I. At $b = 0$ the integrand vanishes, so $I(0) = 0$. Differentiating under the integral (the integrals converge uniformly),

$$I'(b) = \int_0^\infty \frac{\sin br}{r\sqrt{1+r^2}}\,dr, \qquad I'(0) = 0, \tag{B4}$$

$$I''(b) = \int_0^\infty \frac{\cos br}{\sqrt{1+r^2}}\,dr = K_0(b), \tag{B5}$$

where the last integral is a standard representation of the modified Bessel function K_0.

Step 4: integration. Integrating $I'' = K_0$ with $I(0) = I'(0) = 0$, and then integrating by parts,

$$\begin{aligned} I'(b) &= \int_0^b K_0(t)\,dt, \\ I(b) &= \int_0^b \int_0^s K_0(t)\,dt\,ds = \int_0^b (b-t)K_0(t)\,dt \\ &= b\int_0^b K_0(t)\,dt - \int_0^b tK_0(t)\,dt \\ &= b\int_0^b K_0(t)\,dt + bK_1(b) - 1. \end{aligned} \tag{B6}$$

The last line uses $\frac{d}{dt}[tK_1(t)] = -tK_0(t)$ and $tK_1(t) \to 1$ as $t \to 0$, so $-\int_0^b tK_0\,dt = bK_1(b) - 1$.

Step 5: result. With $2k_F = \pi n$,

$$\begin{aligned} e_x(n) &= -\frac{1}{\pi^2} I(\pi n), \\ v_x = \frac{de_x}{dn} &= -\frac{1}{\pi} I'(\pi n) \\ &= -\frac{1}{\pi}\int_0^{\pi n} K_0(t)\,dt. \end{aligned} \tag{B7}$$

Equivalently, e_x is a one-dimensional integral over the Fourier transform of w, which is proportional to K_0; the steps above evaluate it in closed form. Parametrized local-density approximations for this gas, with correlation, are given in Refs. [45, 52]; we use the exchange part only.

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